

DOCKHUB · REQUIREMENTS

DockHub requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/dockhub		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-25	Amish Chadha	Initial scaffold

DockHub requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with riders, hosts or a city, and will be checked by calculation at TRL 3 and revised after co-design sessions (see DKH-PRB-001). Every target is proposed, awaiting Amish.

The **design duty case** used throughout is 20 swaps a day, each returning a SwapCell pack at about 15 % state of charge, in a street location with 4.5 peak sun hours and ambient air between -10 and 45 °C.

Table 1. Requirements. Status is the TRL 2 estimate from DKH-PRC-001.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
R1	Fast swap	60 s or less from tapping the reader to closing the second door, for a rider familiar with the station	Task analysis; later timed trials with riders	Met on estimate (about 30 s)
R2	SwapCell compatibility	Every bay builds to SwapCell interface v0.2: envelope, guide faces, blind-mate receptacle, CAN heartbeat and message set; no change to the interface	Interface review with the SwapCell project	Met by design; one CAN channel per bay needed
R3	Charge speed	20 to 80 % in 1.5 h or less; full charge in 2.5 h or less, per bay, at 25 °C	Charger and pack datasheets; SwapCell charge estimate	Met on estimate (1.2 h and 2.3 h)
R4	Throughput	20 or more swaps a day from 4 bays at the design duty case	Queue and energy calculation	Met on estimate (about 38 a day maximum)
R5	Grid supply	One single-phase branch circuit (120 V 15 A or 230 V 10 A); peak draw 1.5 kW or less	Load calculation	Met on estimate (about 1.25 kW)
R6	Solar contribution	Canopy solar supplies 20 % or more of daily charging energy at the design duty case	Solar yield calculation	Not met (about 13 %)
R7	Fire detection and response	Detect a pack or bay over 60 °C, or smoke, within 10 s; open that bay's charge contactor within 1 s of detection; sound a local alarm and send a remote alert; vent gas to the rear roof louver, away from the user side	Design review; later bench test with a heat source	Met by design on paper; unverified
R8	Fire containment	A thermal runaway in one bay does not spread to a neighboring bay or pack for 30 min or more, and flame does not leave the cabinet front	Later propagation test by an accredited lab	Unverified, at risk
R9	Climate	Enclosure IP54 or better; operate at -10 to 45 °C ambient; never charge a pack outside the pack's reported charge temperature window	Thermal calculation; later chamber test	Not met at 45 °C (interior about 55 °C; charging derates)
R10	Security	1.5 mm or thicker steel; fail-secure bay locks; cabinet anchored to its pad with tamper-resistant fixings; a pack is released only against a valid token and a returned pack	Design review	Met by design
R11	Accessibility and use	All operable parts (reader, buttons, bay handles) 0.38 to 1.22 m (15 to 48 in) above the ground; status shown by light and symbol as well as text; usable with gloves and at night	Model check; later user trials	Met by design (bay handles at about 0.96 m, reader at 0.9 to 1.0 m)
R12	Privacy and data	No cameras or microphones; logs hold pack ID, bay, time, energy and state of health, with the rider as an opaque token; operates offline for 24 h or more and syncs later	Design review	Met by design
R13			Model check	

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
	Sidewalk footprint	Cabinet footprint 1.0 x 0.6 m or less; canopy underside 2.1 m or more above the sidewalk		Met (1.0 x 0.5 m; lowest canopy edge about 2.2 m)
R14	Maintainability	Any charger, lock, bay cradle or controller replaced in 15 min or less by one technician with hand tools, through the service door	Design review	Met by design
R15	Wind and anchoring	Stays upright and intact at a 30 m/s gust with the canopy fitted	Wind load calculation	Met only if anchored (overturning about 3.6 kN·m against about 0.4 kN·m from self-weight)
R16	Cost	Parts 1,200 USD or less for a working prototype, SwapCell packs excluded	Priced BOM (bom/ bom.csv)	Not met with 4 bays fitted (about \$1,423); met with 2 of 4 bays fitted (about \$1,199)

■ Assumptions

- SwapCell pack: 46.8 V nominal, about 468 Wh, charge on a certified 54.6 V 5 A charger, 2.3 h to full and 1.2 h for 20 to 80 %, as in the SwapCell design precis (interface v0.2).
- Charger efficiency about 90 %, cell charge efficiency about 95 %, cabinet auxiliary load (controller, modem, display, fans, lights) about 30 W on average.
- 400 W panel, 4.5 peak sun hours, 0.8 system derate and some clipping at the 5 A bay limit: about 1.3 kWh a day.
- Delivery riders cover roughly 60 to 100 km a day at about 10 Wh/km. This is an assumption to check with riders; it implies 1 to 2 swaps per rider per shift.
- Accessibility reach range from the US ADA forward reach limits (15 to 48 in); local rules may differ.